

2 (**short**) A fluid of density ρ flows through a pipe at velocity V_1 . A contraction is installed in the pipe between sections 1 and 2, as shown in Fig. 2. The contraction is smooth enough to neglect dissipation in it. The jet formed at the subsequent abrupt expansion has uniform velocity and runs parallel to the pipe axis. The pressures p_1 and p_2 are uniform across each section, and the pressure at the back face of the expansion can be assumed to be also p_2 . At section 3, the flow can be considered uniform again.

(a) Can Bernoulli's equation be applied between sections 1 and 2? Show that

$$p_1 - p_2 = \frac{1}{2}\rho V_1^2 K_1,$$

and give the value of the parameter K_1 , which depends only on A_1 and A_2 . [5]

(b) Can Bernoulli's equation be applied between sections 2 and 3? By choosing an appropriate control volume, show that

$$p_3 - p_2 = \frac{1}{2}\rho V_1^2 K_2,$$

and give the value of the parameter K_2 , which depends only on A_1 and A_2 . [5]

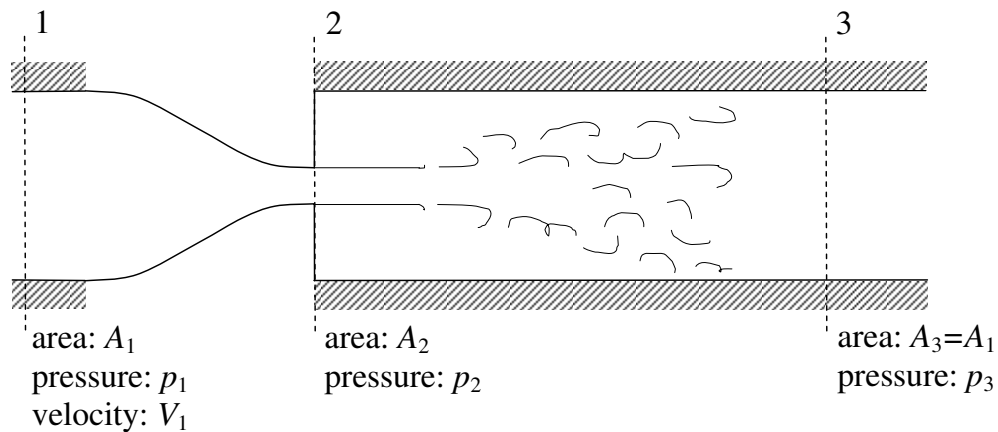


Fig. 2